NON-WSQ COURSE SLIDES

Python Data Analysis Masterclass

Course Code: C193

Trainer: Dr Alfred Ang

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Tertiary Infotech Academy Pte Ltd · UEN 201200696W



00

COURSE ADMINISTRATION

Welcome & Course Orientation

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About the Trainer



Your Trainer

Complete before class

NAME _____

TITLE _____

QUALIFICATIONS _____

EXPERTISE _____

EXPERIENCE _____

CONTACT _____

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal Trainer and Chief Instructional Designer
EXPERTISE	Data analytics, Python, AI and decision support
CERTIFICATION	ACLP-certified adult educator
DELIVERS	Professional technology programmes
EXPERIENCE	Cross-industry analytics and workforce training
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LEARNER INTRODUCTION

Let's Know Each Other

1. **Your work**
   Role and decisions you support.

2. **Python baseline**
   Scripts, notebooks or pandas.

3. **Data challenge**
   One recurring reporting problem.

4. **Outcome**
   What you want to do reliably.

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HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time.

2. **Participate**
   Ask, test and explain.

3. **Respect**
   One conversation at a time.

4. **Protect data**
   Use supplied synthetic data.

5. **Save evidence**
   Retain scripts and outputs.

6. **Learn by doing**
   Run and inspect the code.

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SCHEDULE

Lesson Plan — 2 Days

DAY 1

1. 9:30–11:00 · pandas foundations
2. 11:00–12:45 · Prepare and clean
3. 1:15–3:05 · Transform and summarise
4. 3:05–5:30 · Visualisation labs

DAY 2

1. 9:30–11:25 · Statistics and distributions
2. 11:25–2:50 · Time-series analysis
3. 2:50–5:15 · Advanced pipeline and capstone
4. 5:15–5:30 · Course recap and feedback

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Download Course Material

1 · Sign in

- Open the LMS/TMS portal
- Use your learner account
- Select this course

2 · Download

- Open Courseware
- Download slides and Learner Guide
- Save beside lab folders

3 · Verify

- Check C193
- Use current v1.0
- Ask if a file differs

OPEN LMS / TMS PORTAL

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COURSE REPOSITORY

Access the Python Labs

Clone

- Clone the GitHub repository
- Open labs/Lab-XX
- Run lab.py

Download ZIP

- GitHub > Code > Download ZIP
- Extract the folder tree
- Keep data with each script

Lab discipline

- Use LG for detailed procedures
- Retain outputs as evidence
- All data is synthetic

OPEN COURSE REPOSITORY

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01

TOPIC 01

Data Preparation

pandas foundations · import/export · selection · filtering · cleaning

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Key Concepts — Data Preparation

1

Analytical workflow

Frame the decision, acquire data, prepare, analyse, visualise, interpret and review.

2

Series and DataFrame

Use labelled one- and two-dimensional structures with explicit indexes and data types.

3

Import and export

Read and write CSV, Excel and JSON while preserving schema, provenance and encoding.

4

Selection

Use columns, loc and iloc intentionally; distinguish labels from integer positions.

5

Filtering and slicing

Translate business conditions into reproducible Boolean masks and bounded subsets.

6

Cleaning

Profile missingness, duplicates, invalid types and ranges before applying documented rules.

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Analytics workflow

CONCEPT

1. Frame: Define the decision and grain.
2. Acquire: Record source and provenance.

DECISION / CONTROL

1. Prepare: Profile, clean and validate.
2. Interpret: State finding and limitation.

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Python analytics stack

NumPy

- Arrays and numerical operations.

pandas

- Labelled tables and transformations.

Matplotlib

- Figure and axes control.

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Series and DataFrame

1. **Series**
   One labelled dimension.
2. **DataFrame**
   Rows by columns.
3. **Index**
   Labels used for alignment.
4. **Schema**
   Names, types and rules.

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Import and export

CONCEPT

1. CSV: Portable text; validate encoding.
2. Excel: Sheets and formats.

DECISION / CONTROL

1. JSON: Nested records may need normalisation.
2. Parquet: Typed and compressed.

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Inspect before changing

head / tail

- Preview records.

shape

- Confirm dimensions.

info

- Review types and nulls.

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Select and slice

1. **Columns**
   Select one or many fields.

2. **loc**
   Label-based selection.

3. **iloc**
   Position-based selection.

4. **Slice**
   Retain a bounded subset.

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Filter

CONCEPT

1. Boolean mask: Reproducible conditions.
2. between: Readable numeric bounds.

DECISION / CONTROL

1. isin: Approved category set.
2. Combine: Use parenthesised & | ~.

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Missing data

Detect

- Use isna and notna.

Remove

- Only under a clear rule.

Impute

- Document the chosen method.

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Clean and validate

1. **Duplicates**
   Test declared keys.

2. **Types**
   Convert with controlled errors.

3. **Ranges**
   Expose impossible values.

4. **Reconcile**
   Explain every changed total.

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Analytics, reporting, BI and data science

1. **Reporting**
   Describes agreed metrics from known data.

2. **Business intelligence**
   Explores patterns through governed dashboards.

3. **Analytics**
   Tests questions and explains decision-relevant variation.

4. **Data science**
   Builds and evaluates models or data products.

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Decision-first analytical framing

1

Decision: What action could change?

2

Question: What must be learned?

3

Evidence: Which measure and comparison answer it?

4

Guardrail: What must not worsen?

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Observational unit and data grain

FOUNDATION

1. Unit: The entity or event represented by one observation.
2. Grain: The exact level of detail in each row.

INTERPRETATION / CONTROL

1. Key: Fields expected to identify one row at that grain.
2. Measure: A value meaningful at the declared grain.

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Data quality dimensions

1. **Completeness**
   Required values are present.

2. **Validity**
   Values follow type, range and format rules.

3. **Uniqueness**
   Declared entities are not duplicated.

4. **Consistency**
   Values agree across fields and sources.

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Missing data is an analytical signal

FOUNDATION

1. MCAR: Missingness is unrelated to observed or unobserved values.
2. MAR: Missingness can be explained by observed variables.

INTERPRETATION / CONTROL

1. MNAR: Missingness relates to the unseen value itself.
2. Response: Drop, impute, flag or redesign with documented impact.

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Duplicate records and business keys

1

Exact duplicate: Every field repeats.

2

Key duplicate: Declared identifier repeats.

3

Legitimate repeat: Same entity, different event or time.

4

Resolution: Keep, aggregate or reject by rule.

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Types, categories and encoding

1

Numeric

Supports meaningful arithmetic.

2

Categorical

Represents a controlled set of labels.

3

Datetime

Preserves calendar and interval semantics.

4

Boolean

Stores explicit true or false states.

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Reconciliation and an audit trail

FOUNDATION

1. Before: Record row count, key count and control totals.
2. Change: Name the rule and affected records.

INTERPRETATION / CONTROL

1. After: Recalculate the same controls.
2. Explain: Account for every difference and exception.

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Population, sample and selection bias

1

Population

All units the claim is intended to cover.

2

Sample

Observed units used in the analysis.

3

Sampling frame

Operational list from which units can enter.

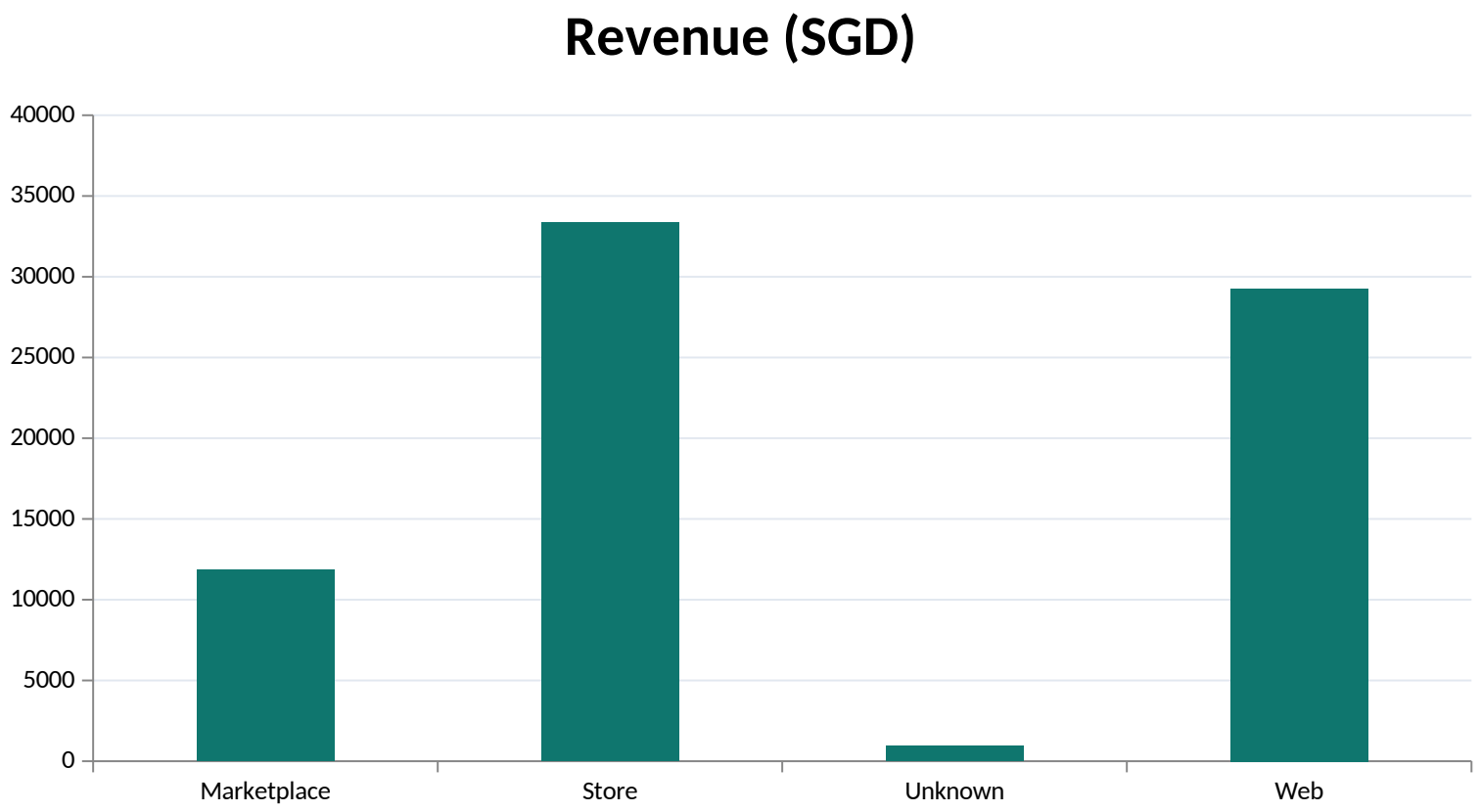
4

Selection bias

Inclusion relates to the outcome or exposure.

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Clean revenue by channel (SGD)



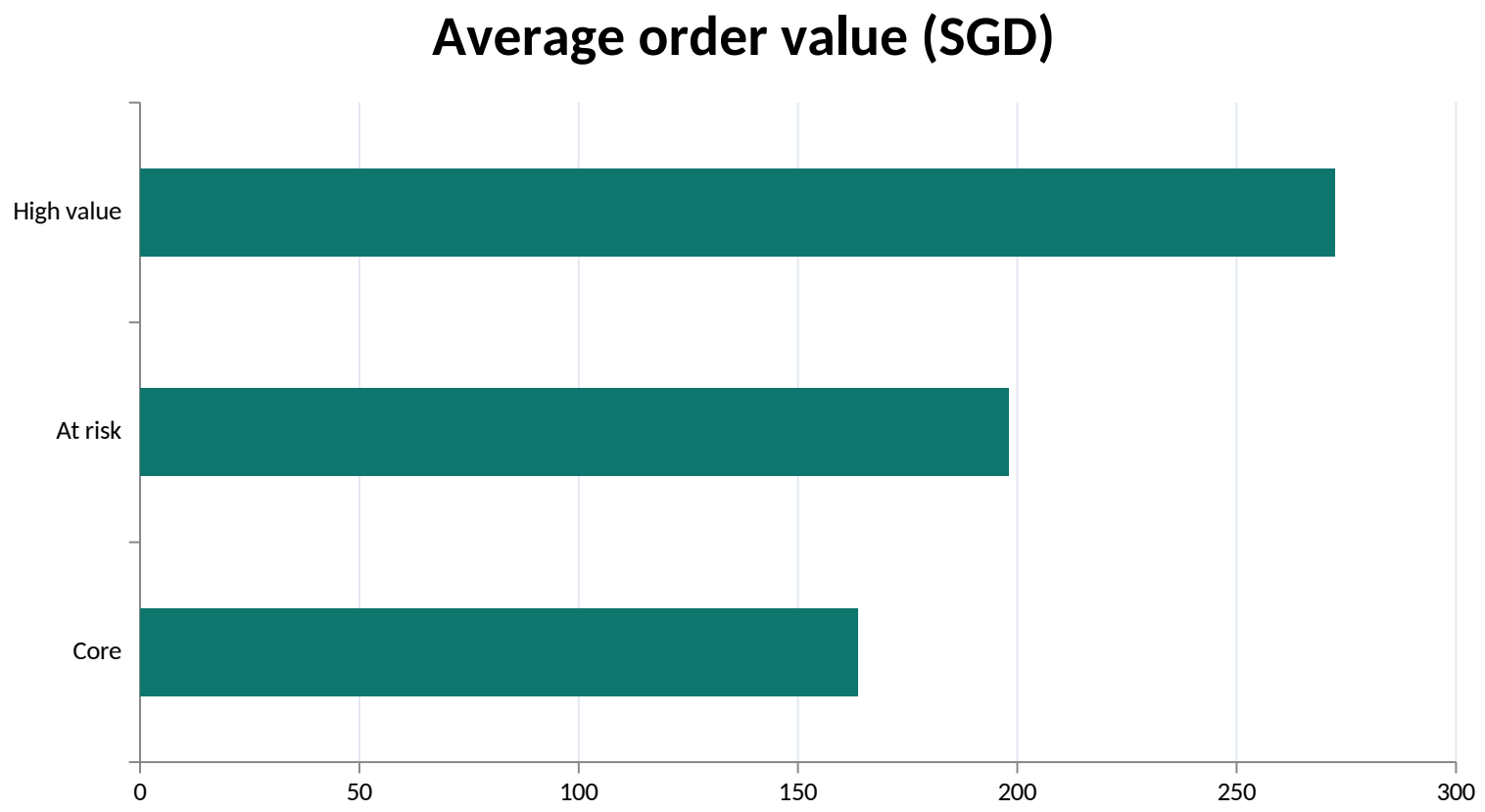
PYTHON → PPTX

A cleaned table should reconcile before it is visualised.

Editable native PowerPoint chart derived from pandas output.

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Customer segment value (SGD)



PYTHON → PPTX

A zero baseline keeps segment magnitude honest.

Editable native PowerPoint chart derived from pandas output.

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LAB 01 · CONCEPT BRIEF

Profile and Clean Retail Transactions

Audit a synthetic omnichannel retail extract, repair missing and invalid values, remove duplicate records and preserve an exception log.

YOU WILL BUILD

A clean retail DataFrame, data-quality summary and before/after revenue comparison chart.

TOOLS Python, pandas, Matplotlib



ACCEPTANCE TEST

The script runs without error, clean_data.csv has unique order IDs, critical fields are complete and the quality report reconciles every change.

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LAB 02 · CONCEPT BRIEF

Select, Filter and Segment Customer Orders

Use label-based and position-based selection, Boolean filters and slicing to answer a realistic customer-retention question.

YOU WILL BUILD

Reusable customer segments and a chart comparing average order value by segment.

TOOLS Python, pandas, Matplotlib



ACCEPTANCE TEST

The loc and iloc samples are saved, every segment rule is reproducible and the chart uses an honest zero baseline.

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TOPIC RECAP

Recap — Data Preparation

1. **Knowledge**
   Analytical workflow and related principles.

2. **Ability**
   LO1

3. **Evidence**
   Runnable Python and pandas outputs.

4. **Next**
   Connect to the next analytical layer.

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02

TOPIC 02

Data Transformation

concat · merge · sort · groupby · aggregation · pivot tables

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Key Concepts — Data Transformation

1

Concatenate

Append compatible records by row or combine aligned features by column.

2

Merge

Join tables on validated keys and inspect cardinality, unmatched records and duplicated rows.

3

Sort

Order data by one or more fields without confusing row order with analytical importance.

4

GroupBy

Split data into meaningful segments, apply measures and combine the results.

5

Aggregation

Use named aggregations so each metric, denominator and output column is explicit.

6

Pivot

Reshape long and wide data for comparison while retaining the underlying grain.

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Concatenate

CONCEPT

1. Rows: Append compatible extracts.
2. Columns: Combine aligned features.

DECISION / CONTROL

1. Index: Labels control alignment.
2. Schema: Standardise before concat.

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Merge

Left

- Keep the analytical base.

Inner

- Keep matched records only.

Outer

- Expose all exceptions.

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Join cardinality

1. **One-to-one**
   Unique keys on both sides.

2. **Many-to-one**
   Transactions to master data.

3. **One-to-many**
   Entity to events.

4. **Many-to-many**
   May multiply rows.

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Key quality

CONCEPT

1. Complete: Required keys populated.
2. Consistent: Case, spaces and type agree.

DECISION / CONTROL

1. Unique: Reference keys meet contract.
2. Stable: Meaning does not drift.

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Sort

Values

- Order by one or more fields.

Direction

- Ascending can vary by field.

Missing

- Choose visible placement.

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GroupBy

1. **Split**
   Partition by category.

2. **Apply**
   Calculate group measures.

3. **Combine**
   Return a summary.

4. **Grain**
   Know each result row.

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Named aggregation

CONCEPT

1. Count: Population size.
2. Mean/median: Typical value.

DECISION / CONTROL

1. Rate: Make denominator explicit.
2. Names: Use decision-ready outputs.

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Pivot table

Index

- Primary row grouping.

Columns

- Comparison dimension.

Values

- Measure to aggregate.

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Long and wide

1. **Long**
   One observation per row.
2. **Wide**
   Category per column.
3. **melt**
   Wide to long.
4. **pivot**
   Long to wide.

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Tidy data and long-form structure

1

Variable

Each measured attribute has its own column.

2

Observation

Each unit at the declared grain has one row.

3

Value

Each cell contains one value.

4

Table

Each observational unit has a coherent table.

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Choose a join from the business question

1

Base: Which table defines the required population?

2

Match: Which key expresses the relationship?

3

Join: Left, inner or outer based on the decision.

4

Audit: Inspect matched and unmatched records.

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Join cardinality and row multiplication

FOUNDATION

1. 1 × 1: One key on each side produces at most one row.
2. n × 1: Many transactions map to one reference record.

INTERPRETATION / CONTROL

1. 1 × n: One entity expands into several valid events.
2. n × n: Repeated keys on both sides can multiply rows.

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Unmatched-record audit

1. **Left only**
   Base records with no reference match.

2. **Right only**
   Reference records never used by the base.

3. **Both**
   Successfully matched records.

4. **Disposition**
   Correct, enrich, exclude or escalate by rule.

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Aggregation grain and denominator

FOUNDATION

1. Group: Dimensions defining each result row.
2. Measure: Count, sum, mean, median or rate.

INTERPRETATION / CONTROL

1. Denominator: Eligible observations behind a rate or mean.
2. Reconcile: Group results must explain the base population.

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Weighted and unweighted averages

FORMULA / WORKED VALUE

$$\bar{x}_w = \Sigma(w_i x_i) / \Sigma w_i$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Unweighted:** Every group mean receives equal influence.
2. **Weighted:** Influence follows an exposure such as students or sales.
3. **Weight:** Must reflect the decision and be non-negative.
4. **Check:** Report total weight and compare both results.

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Rates, proportions and explicit bases

FOUNDATION

1. Numerator: Events meeting the stated condition.
2. Denominator: Eligible opportunities for that event.

INTERPRETATION / CONTROL

1. Period: Time window shared by numerator and base.
2. Exclusions: Missing or ineligible records disclosed.

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GroupBy, pivot table and crosstab

1. **GroupBy**
   Flexible split-apply-combine calculations.

2. **Pivot table**
   Matrix summary with explicit aggregation.

3. **Crosstab**
   Counts or proportions for categorical combinations.

4. **Reshape**
   melt and pivot change layout, not meaning.

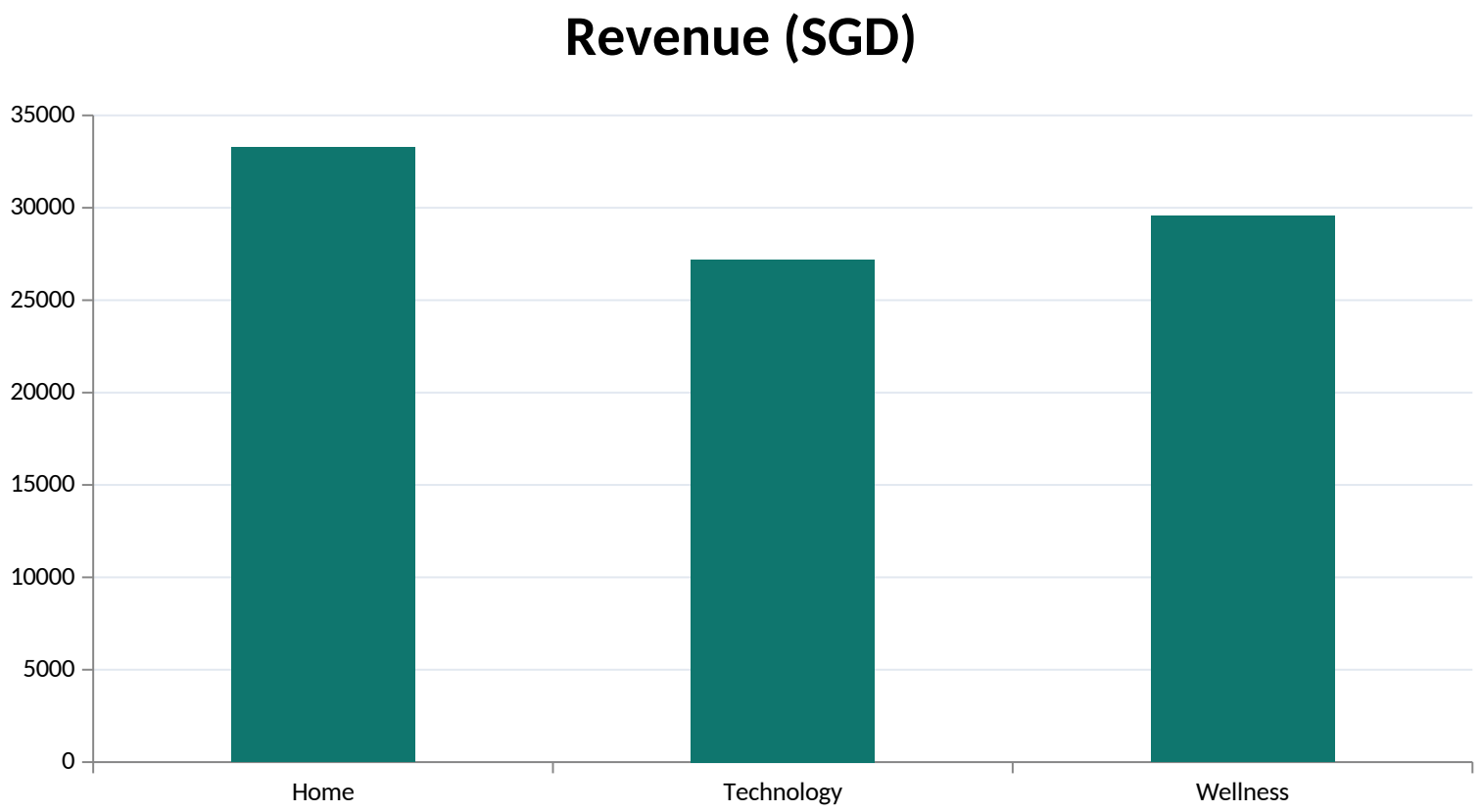
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Transformation pipeline controls

1. Contract: Assert required columns and types.
2. Transform: Use named, pure stages.
3. Validate: Check grain, ranges and reconciliations.
4. Publish: Save outputs with provenance and limitations.

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Revenue after dataset integration (SGD)



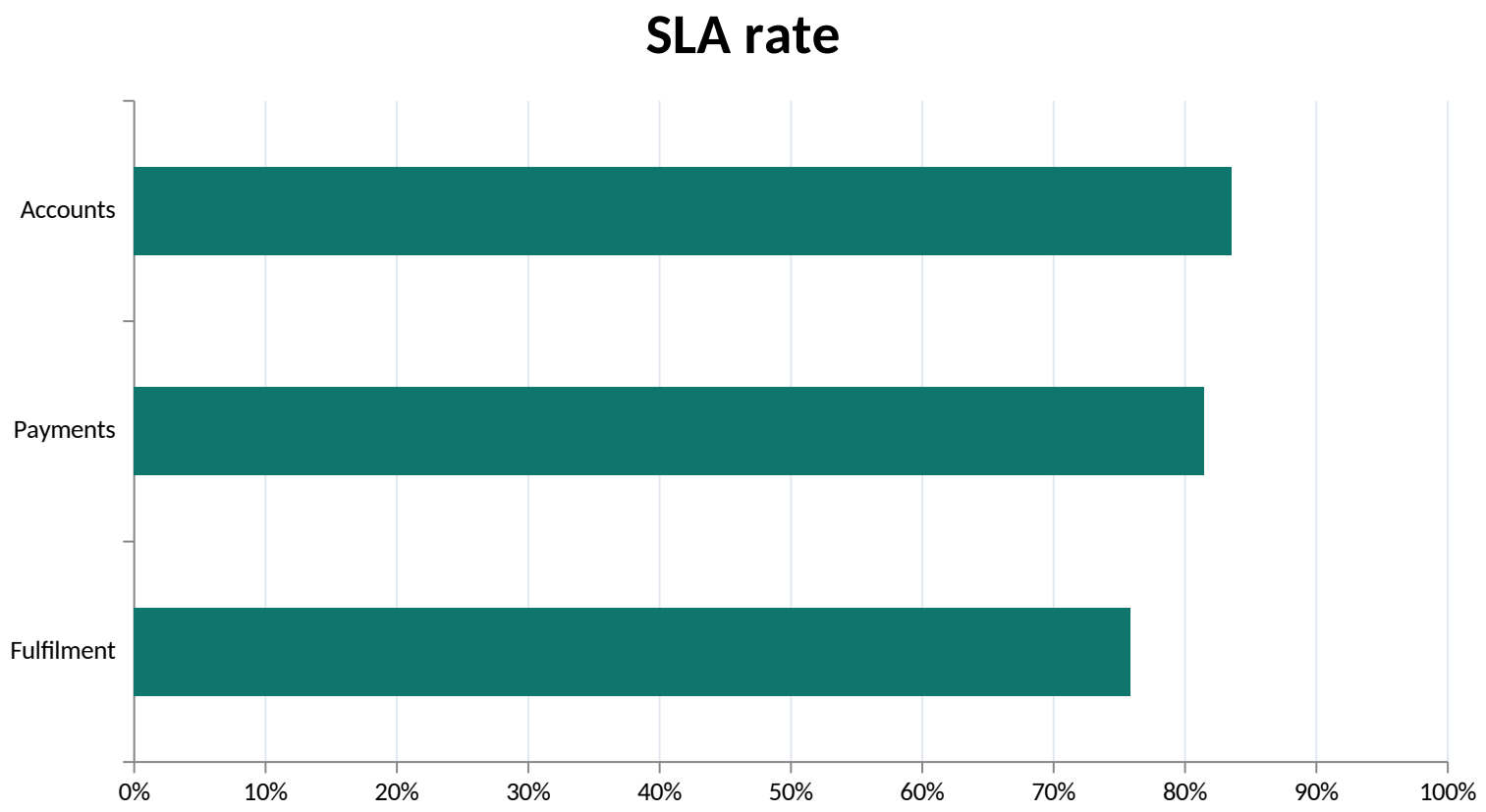
PYTHON → PPTX

Validated joins make the aggregated result defensible.

Editable native PowerPoint chart derived from pandas output.

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SLA rate by support team



PYTHON → PPTX

Rate and workload answer different management questions.

Editable native PowerPoint chart derived from pandas output.

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LAB 03 · CONCEPT BRIEF

Integrate Orders, Products and Targets

Join transaction, product and monthly-target tables with validated keys, cardinality checks and unmatched-record evidence.

YOU WILL BUILD

An enriched order table, join audit and product-category performance chart.

TOOLS Python, pandas merge, concat



ACCEPTANCE TEST

Row counts reconcile, the many-to-one validations pass, unmatched keys are reported and totals before/after integration are explained.

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LAB 04 · CONCEPT BRIEF

Summarise Service Operations with GroupBy and Pivot

Transform service-ticket data into SLA and workload summaries using named aggregations and pivot tables.

YOU WILL BUILD

A tidy SLA scorecard, priority-by-team pivot and stacked workload chart.

TOOLS Python, pandas groupby, agg, pivot_table



ACCEPTANCE TEST

The pivot total reconciles, each metric has an explicit denominator and the SLA result is not confused with ticket volume.

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TOPIC RECAP

Recap — Data Transformation

1. **Knowledge**
   Concatenate and related principles.

2. **Ability**
   LO2

3. **Evidence**
   Runnable Python and pandas outputs.

4. **Next**
   Connect to the next analytical layer.

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03

TOPIC 03

Data Visualization

pandas plots · Matplotlib · Seaborn · chart choice · visual integrity

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Key Concepts — Data Visualization

1

Question first

Choose the chart from the comparison, trend, distribution or relationship question.

2

Matplotlib

Control the figure, axes, labels, annotations and export for reproducible publication.

3

Seaborn

Create statistical graphics with semantic mappings and consistent themes.

4

Comparison and trend

Use bars for category magnitude and lines for ordered time.

5

Distribution

Use histograms and box plots to reveal shape, spread and candidate outliers.

6

Relationship

Use scatter, regression, pair plots and heatmaps without implying causation.

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Choose charts by question

CONCEPT

1. Trend: Line across ordered time.
2. Comparison: Bar across categories.

DECISION / CONTROL

1. Distribution: Histogram or box plot.
2. Relationship: Scatter plot.

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pandas and Matplotlib

DataFrame.plot

- Fast first view.

Figure

- Full visual canvas.

Axes

- Coordinate system.

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Line charts

1. **Ordered x**
   Use time or sequence.
2. **Intervals**
   Represent spacing truthfully.
3. **Series**
   Limit competing lines.
4. **Annotations**
   Call out turning points.

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Bar charts

CONCEPT

1. Length: Encodes magnitude.
2. Zero: Required baseline.

DECISION / CONTROL

1. Sort: Use when rank matters.
2. Horizontal: Helps long labels.

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Composition charts

Stacked bar

- Total plus contribution.

Area

- Accumulated trend.

Pie

- Only a few part-to-whole slices.

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Distribution charts

1. **Histogram**
   Shape depends on bins.

2. **Box plot**
   Median and IQR.

3. **Density**
   Smoothed distribution.

4. **Outlier**
   Candidate for investigation.

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Relationship charts

CONCEPT

1. Scatter: Two numerical variables.
2. Regression: Estimated linear trend.

DECISION / CONTROL

1. Pair plot: Screen several variables.
2. Causality: Pattern is not proof.

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Heatmap

Matrix

- Colour encodes cells.

Correlation

- Use -1 to +1 centred scale.

Labels

- Avoid false precision.

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Subplots

1. **Small multiples**
   One panel per segment.
2. **Shared scale**
   Enables comparison.
3. **Layout**
   Preserve readable labels.
4. **Legend**
   Reduce visual search.

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Visual integrity

CONCEPT

1. Axes: Truthful units and limits.
2. Colour: Accessible and restrained.

DECISION / CONTROL

1. Context: Period, source, denominator.
2. Uncertainty: Show material limitations.

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Pre-attentive attributes guide attention

1. **Position**
   Supports precise comparison along a common scale.

2. **Length**
   Makes magnitude easy to compare.

3. **Colour**
   Highlights category or exception when used sparingly.

4. **Size**
   Attracts attention but is less precise than length.

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Chart selection decision tree

1

Trend: Line for values across ordered time.

2

Compare: Bar or dot plot for category magnitude.

3

Distribution: Histogram, density or box plot.

4

Relationship: Scatter plot for two numeric variables.

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Visual encoding accuracy

FOUNDATION

1. Position: Most accurately compared on a common scale.
2. Length: Strong for bars sharing a baseline.

INTERPRETATION / CONTROL

1. Angle and area: Harder to compare precisely.
2. Colour intensity: Best for ordered patterns, not exact values.

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Zero baselines and honest axes

FOUNDATION

1. Bars: Start at zero because length encodes magnitude.
2. Lines: May use a focused range when disclosed and justified.

INTERPRETATION / CONTROL

1. Intervals: Use truthful spacing and consistent time units.
2. Breaks: Mark discontinuities instead of hiding them.

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Colour accessibility and semantics

1

Categorical

Use distinct hues for a small set of groups.

2

Sequential

Use ordered light-to-dark for magnitude.

3

Diverging

Centre two directions on a meaningful midpoint.

4

Accessible

Check contrast and avoid colour-only meaning.

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Histogram bins change the story

FOUNDATION

1. Too few: Important modes and gaps disappear.
2. Too many: Random noise appears as structure.

INTERPRETATION / CONTROL

1. Comparison: Use shared boundaries across groups.
2. Disclosure: State bin width or count and sample size.

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Box plot anatomy and limitations

1. **Median**
   Line at the 50th percentile.

2. **Box**
   Q1 to Q3, the middle 50 percent.

3. **Whiskers**
   Often extend to 1.5 IQR from the quartiles.

4. **Fliers**
   Candidates for investigation, not automatic errors.

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Scatter plots, overplotting and trend

FOUNDATION

1. Pair: Each point must represent one matched observation.
2. Overlap: Use transparency, jitter or aggregation when dense.

INTERPRETATION / CONTROL

1. Trend: Show a fitted line only with a clear descriptive purpose.
2. Causality: Association alone cannot identify an intervention effect.

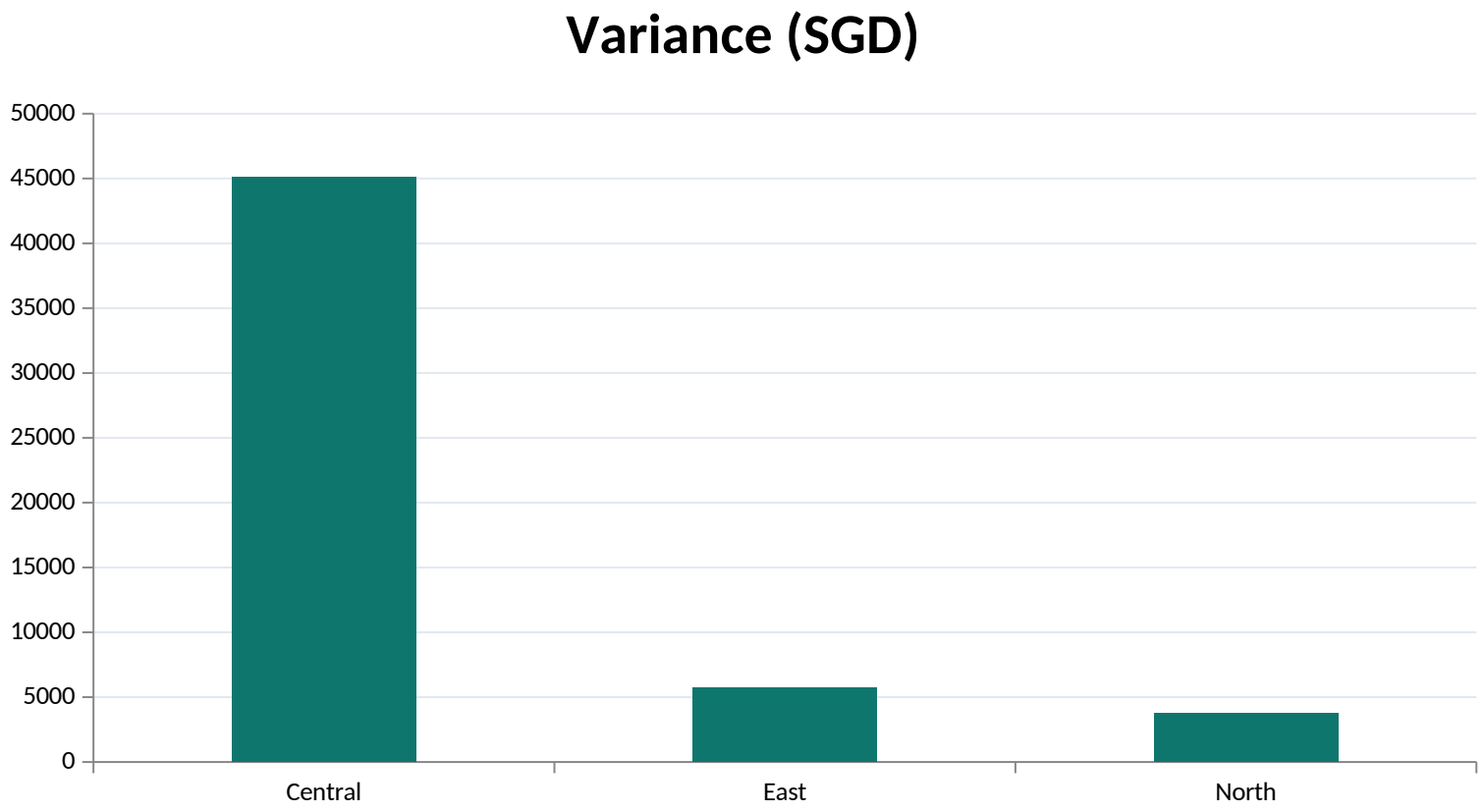
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Decision-ready annotation and context

1. Message: State the supported finding in the title.
2. Evidence: Show units, period, base and comparison.
3. Annotation: Label the turning point or exception.
4. Caveat: Disclose uncertainty, exclusions and source.

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Revenue variance to target (SGD)



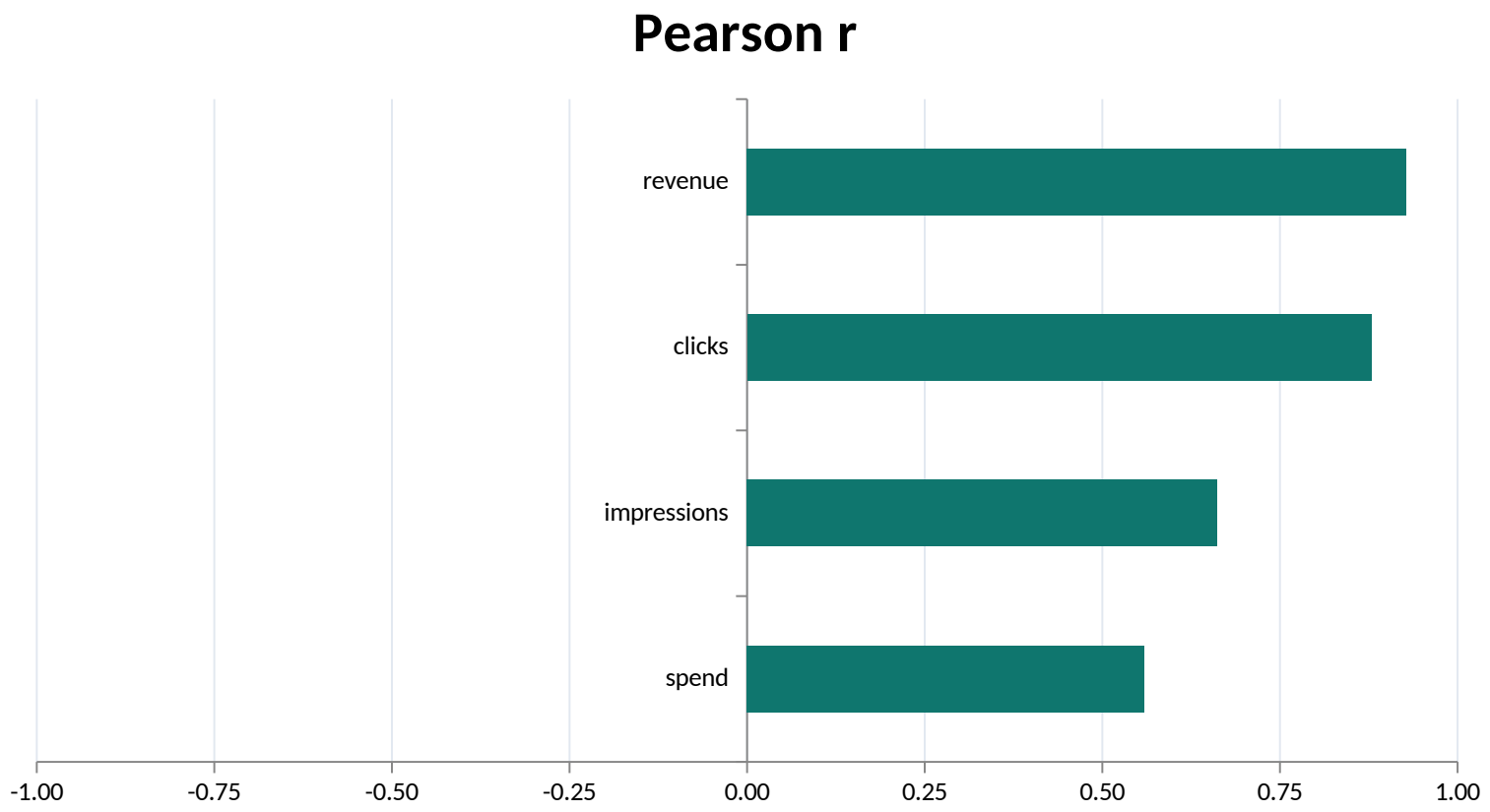
PYTHON → PPTX

Direction and size of variance focus attention on action.

Editable native PowerPoint chart derived from pandas output.

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Correlation with conversions



PYTHON → PPTX

Association is not causal proof.

Editable native PowerPoint chart derived from pandas output.

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LAB 05 · CONCEPT BRIEF

Visualise Monthly Retail Performance

Select chart types for time trend, category comparison and target variance, then annotate the decision signal.

YOU WILL BUILD

A three-chart management pack: monthly trend, regional comparison and variance waterfall-style bar chart.

TOOLS Python, pandas plot, Matplotlib



ACCEPTANCE TEST

All three charts render, labels and units are complete, bars use a zero baseline and the rationale connects chart choice to the question.

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LAB 06 · CONCEPT BRIEF

Explore Marketing Relationships and Correlation

Use scatter, regression, pair plots and a correlation heatmap to explore marketing performance without causal overclaim.

YOU WILL BUILD

A relationship-analysis pack with scatter/regression plot, pair plot, heatmap and evidence statement.

TOOLS Python, pandas, Matplotlib, Seaborn



ACCEPTANCE TEST

The graphics render, correlation values are reproducible and the interpretation explicitly separates association from causation.

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TOPIC RECAP

Recap — Data Visualization

1. **Knowledge**
   Question first and related principles.

2. **Ability**
   LO3

3. **Evidence**
   Runnable Python and pandas outputs.

4. **Next**
   Connect to the next analytical layer.

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04

TOPIC 04

Data Analysis

data types · descriptive statistics · covariance · correlation · time series

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Key Concepts — Data Analysis

1

Data types

Distinguish categorical and numerical data, including nominal, ordinal, discrete and continuous.

2

Centre

Use mean, median and mode according to distribution and decision context.

3

Spread

Use range, variance, standard deviation and quartiles to quantify variability.

4

Covariance

Measure whether variables move together, while recognising scale dependence.

5

Correlation

Standardise linear association from -1 to +1; never treat association as causation.

6

Time series

Parse datetime, index by time, resample, roll and compare periods without leakage.

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Types of data

CONCEPT

1. Nominal: Unordered labels.
2. Ordinal: Ordered categories.

DECISION / CONTROL

1. Discrete: Countable numbers.
2. Continuous: Measured interval values.

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Centre

Mean

- Sensitive to extremes.

Median

- Robust midpoint.

Mode

- Most frequent value.

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Spread

1. **Range**
   Maximum minus minimum.

2. **Variance**
   Average squared distance.

3. **Standard deviation**
   Spread in original unit.

4. **IQR**
   Middle 50 percent.

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Descriptive statistics

CONCEPT

1. describe: Standard summary.
2. agg: Selected named measures.

DECISION / CONTROL

1. Group: Compare consistent metrics.
2. Missing: State denominator.

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Categorical analysis

value_counts

- Frequency by category.

normalize

- Convert to proportion.

crosstab

- Joint category counts.

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Covariance and correlation

1. **Covariance**
   Scale-dependent co-movement.

2. **Pearson**
   Linear association.

3. **Spearman**
   Rank association.

4. **Caution**
   Neither proves causality.

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Datetime

CONCEPT

1. to_datetime: Parse strings.
2. Format: Specify known patterns.

DECISION / CONTROL

1. Timezone: Localise deliberately.
2. Invalid: Coerce and audit.

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DatetimeIndex

Slice

- Select dates and periods.

Properties

- Year, month, weekday, hour.

Order

- Sort and enforce uniqueness.

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Resample

1. **Downsample**
   Hourly to daily.

2. **Upsample**
   Create finer intervals.

3. **Rule**
   D, W, M and more.

4. **Measure**
   Mean, max, sum or last.

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Rolling and peaks

CONCEPT

1. Trailing: Current and past only.
2. Window: Count or time offset.

DECISION / CONTROL

1. Threshold: Limit or percentile.
2. Evidence: Retain peak records.

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Population, sample, parameter and statistic

1. **Population**
   Full group about which a claim is intended.

2. **Sample**
   Observed subset used for calculation.

3. **Parameter**
   Unknown population quantity such as μ or σ.

4. **Statistic**
   Sample estimate such as $\bar{x}$ or s.

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Measurement scales determine valid analysis

1. **Nominal**
   Labels with no inherent order.

2. **Ordinal**
   Ordered labels with unknown spacing.

3. **Interval**
   Equal differences without a true zero.

4. **Ratio**
   Equal differences and a meaningful zero.

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Mean, median and mode under an outlier

FORMULA / WORKED VALUE

$$\bar{x} = \Sigma x_i / n$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Data:** 12, 13, 13, 14, 48
2. **Mean:** 20; pulled upward by 48.
3. **Median:** 13; robust midpoint.
4. **Mode:** 13; most frequent observed value.

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Quartiles, IQR and the five-number summary

FORMULA / WORKED VALUE

$$IQR = Q3 - Q1$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Minimum:** Smallest observed value.
2. **Q1:** 25th percentile.
3. **Median:** 50th percentile.
4. **Q3 / maximum:** 75th percentile and largest value.

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Variance and standard deviation

FORMULA / WORKED VALUE

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$
$$s = \sqrt{s^2}$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Deviation:** Distance of each value from the mean.
2. **Square:** Prevents positive and negative distances cancelling.
3. **Variance:** Average squared deviation using n−1 for a sample.
4. **SD:** Square root returns to the original unit.

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Coefficient of variation

FORMULA / WORKED VALUE

$$CV = s / \bar{x} \times 100\%$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Purpose:** Compares relative spread across different scales.
2. **Numerator:** Standard deviation.
3. **Denominator:** Mean of the same measure.
4. **Caution:** Unstable when the mean is near zero or can be negative.

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Z-scores and unusual observations

FORMULA / WORKED VALUE

$$z = (x - \bar{x}) / s$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Centre:** Subtract the reference mean.
2. **Scale:** Divide by the reference standard deviation.
3. **Meaning:** Number of standard deviations from the mean.
4. **Caution:** A large z-score is unusual, not automatically wrong.

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Covariance and correlation

FORMULA / WORKED VALUE

$$r = \mathrm{cov}(X,Y) / (s_X s_Y)$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **Covariance:** Direction of joint movement in product units.
2. **Correlation:** Standardised association from −1 to +1.
3. **Sign:** Shows direction, not importance or causality.
4. **Magnitude:** Shows linear strength and is sensitive to outliers.

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Pearson or Spearman correlation

FOUNDATION

1. Pearson: Linear association between numerical variables.
2. Spearman: Monotonic association based on ranks.

INTERPRETATION / CONTROL

1. Inspect: Plot form, outliers, ties and subgroup patterns.
2. Report: Method, coefficient, sample size and limitation.

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Time-series components

1. **Level**
   Typical magnitude around which a series varies.

2. **Trend**
   Persistent long-run movement.

3. **Seasonality**
   Pattern repeating at a known calendar frequency.

4. **Irregular**
   Residual shocks, noise and anomalies.

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Resampling, rolling windows and leakage

1

Parse: Create valid, timezone-aware timestamps.

2

Order: Sort and resolve duplicate intervals.

3

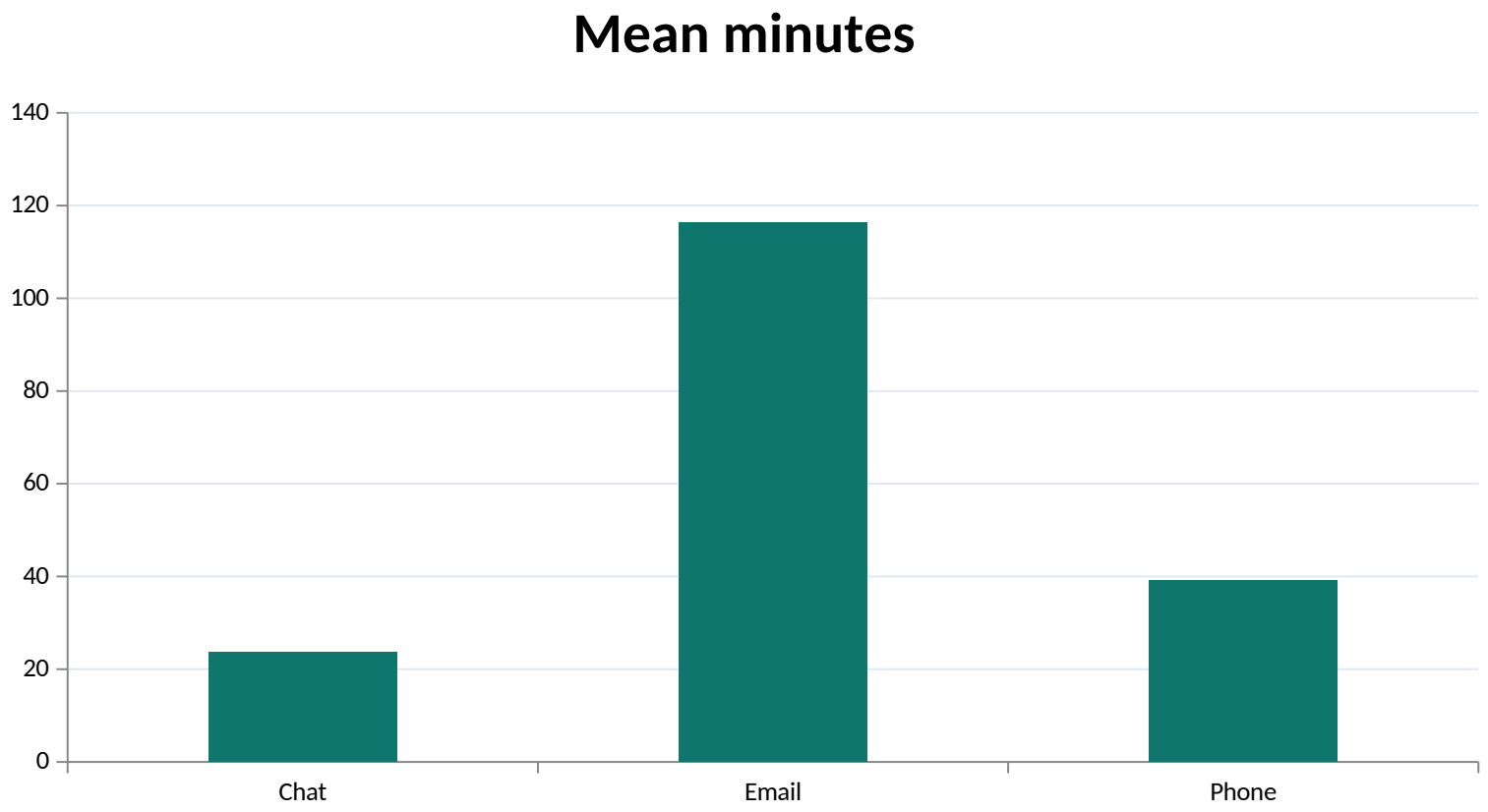
Resample: Aggregate to the decision frequency.

4

Roll: Use only information available at prediction time.

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Mean resolution time by channel



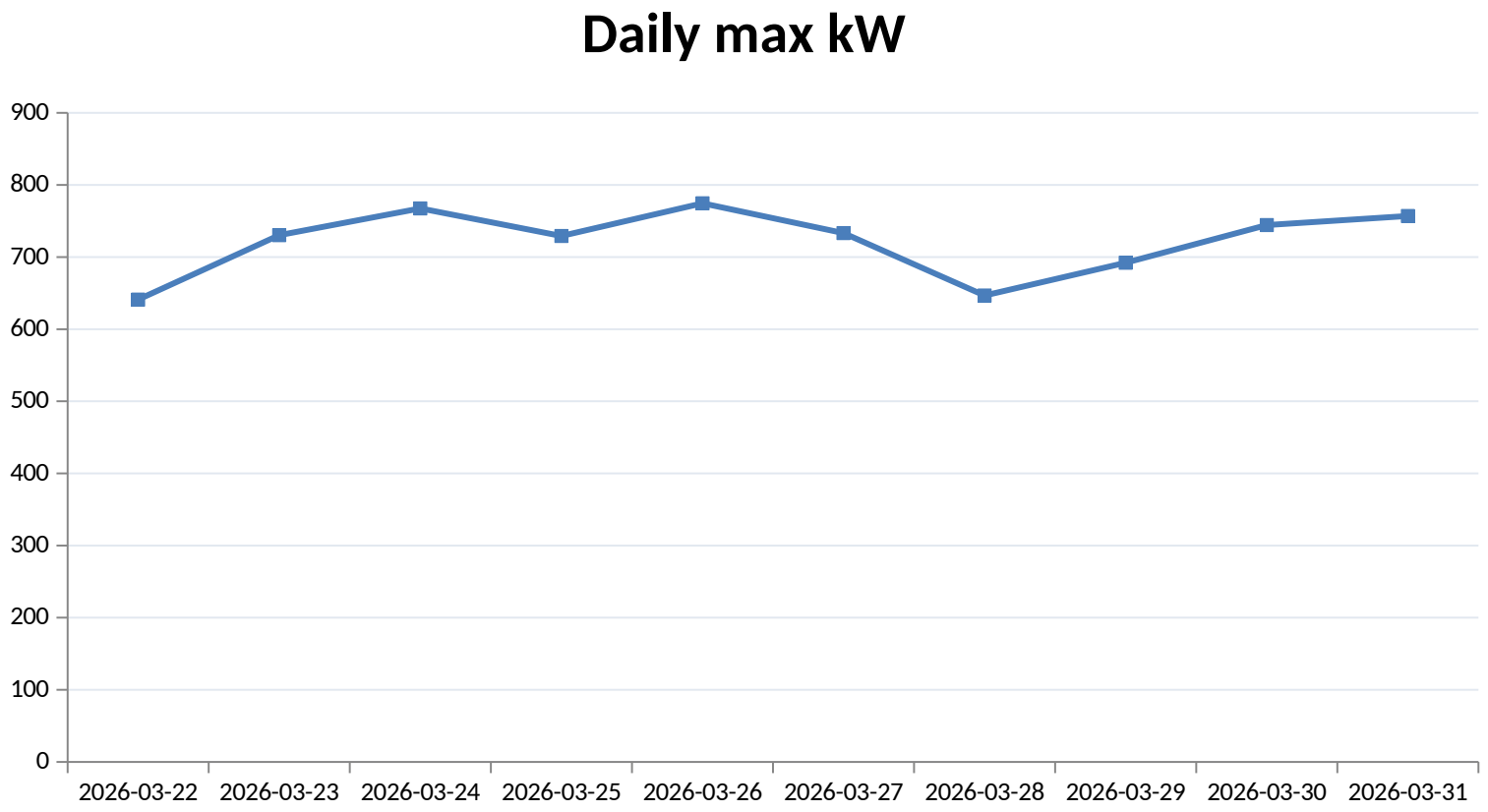
PYTHON → PPTX

Compare mean with median when the distribution is skewed.

Editable native PowerPoint chart derived from pandas output.

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Recent peak energy demand



PYTHON → PPTX

Trailing monitoring must not use future values.

Editable native PowerPoint chart derived from pandas output.

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LAB 07 · CONCEPT BRIEF

Analyse Service Quality Distributions

Compare centre and spread across service channels and identify candidate outliers without deleting them automatically.

YOU WILL BUILD

Descriptive-statistics table, histogram, box plot and evidence-based channel comparison.

TOOLS Python, pandas, Matplotlib, Seaborn



ACCEPTANCE TEST

Statistics reconcile to the source, candidate outliers remain traceable and the interpretation uses median when the distribution is skewed.

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LAB 08 · CONCEPT BRIEF

Resample and Monitor Energy Demand

Parse an hourly time series, resample it to daily and weekly frequency, calculate rolling demand and detect peaks without future leakage.

YOU WILL BUILD

Daily and weekly energy summaries, rolling-demand chart and a peak-demand exception table.

TOOLS Python, pandas DatetimeIndex, resample, rolling, Matplotlib



ACCEPTANCE TEST

The index is complete and sorted, resampled totals are explainable, rolling calculations use no future values and peak days are retained as evidence.

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TOPIC RECAP

Recap — Data Analysis

1

Knowledge

Data types and related principles.

2

Ability

LO4

3

Evidence

Runnable Python and pandas outputs.

4

Next

Connect to the next analytical layer.

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05

TOPIC 05

Advanced Data Analytics

pipe · apply · custom functions · linear regression · credibility review

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Key Concepts — Advanced Data Analytics

1

Method chains

Compose readable transformations with assign and pipe instead of hidden intermediate edits.

2

Apply and vectorisation

Prefer vectorised operations; use apply for justified custom row or column logic.

3

Regression

Estimate a linear relationship between an outcome and one or more predictors.

4

Residuals and MSE

Inspect unexplained errors and quantify average squared prediction loss.

5

Interpretation

Separate coefficient direction, model fit, uncertainty and practical meaning.

6

Credibility review

Check data quality, assumptions, leakage, limitations and reproducibility before release.

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Method chains

CONCEPT

1. Readable: Follow analytical sequence.
2. Reproducible: No hidden manual state.

DECISION / CONTROL

1. Testable: One function per responsibility.
2. Reviewable: Changes remain traceable.

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pipe and assign

pipe

- Pass data into a function.

assign

- Create features in a chain.

Return

- Each function returns the next table.

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Vectorisation and apply

1. **Vectorise**
   Fast native array operations.

2. **apply**
   Flexible custom logic.

3. **Cost**
   Row-wise apply can be slow.

4. **Choice**
   Prefer clarity and scale.

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Custom functions

CONCEPT

1. Input: Expected schema.
2. Transform: No hidden mutation.

DECISION / CONTROL

1. Output: Defined grain and fields.
2. Errors: Fail loudly and clearly.

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Linear regression

Outcome Y

- Value to explain or predict.

Predictors X

- Inputs available at decision time.

Coefficient

- Conditional estimated change.

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Train and test

1. **Train**
   Fit model parameters.

2. **Test**
   Estimate unseen performance.

3. **Random state**
   Reproducible split.

4. **Leakage**
   No future or test information.

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Residuals and MSE

CONCEPT

1. Residual: Actual minus predicted.
2. Pattern: Signals model weakness.

DECISION / CONTROL

1. MSE: Average squared loss.
2. RMSE: Returns to outcome units.

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R-squared

Meaning

- Variation explained.

Test R2

- Generalisation evidence.

Negative

- Worse than mean baseline.

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Strengths and weaknesses

1. **Transparent**
   Interpretable baseline.

2. **Fast**
   Efficient to fit.

3. **Linearity**
   May miss curves.

4. **Outliers**
   Can influence fit.

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Credibility review

CONCEPT

1. Data: Relevant and representative.
2. Method: Assumptions fit design.

DECISION / CONTROL

1. Evidence: Diagnostics and held-out metrics.
2. Claim: Stay within what data supports.

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Four levels of analytics

1. Descriptive: What happened?
2. Diagnostic: What factors are associated with it?
3. Predictive: What may happen under similar conditions?
4. Prescriptive: Which action best meets an objective and constraints?

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Correlation, regression and causality

FOUNDATION

1. Correlation: Symmetric description of association.
2. Regression: Conditional outcome estimate given predictors.

INTERPRETATION / CONTROL

1. Prediction: Performance on unseen cases under similar conditions.
2. Causal effect: Change attributable to an intervention under a design.

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Simple linear regression model

FORMULA / WORKED VALUE

$$\hat{y} = \beta_0 + \beta_1 x$$

$$e = y - \hat{y}$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **$\hat{y}$:** Predicted outcome for a given x.
2. **β_0:** Predicted outcome when x equals zero.
3. **β_1:** Estimated change in outcome for one-unit increase in x.
4. **e:** Observed minus predicted outcome.

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Interpreting coefficients responsibly

1. **Direction**
   Positive or negative conditional association.

2. **Magnitude**
   Outcome-unit change per predictor unit.

3. **Reference**
   Baseline or omitted category for categorical inputs.

4. **Scope**
   Applies to the fitted population and observed range.

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Train, test and overfitting

1. Split: Reserve unseen observations before fitting.
2. Train: Estimate parameters and tune choices.
3. Test: Evaluate once on held-out data.
4. Compare: Check generalisation against a simple baseline.

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Residual diagnostics

1. **Residual versus fitted**
   Reveals curvature or changing spread.

2. **Distribution**
   Shows skew and extreme errors.

3. **Time order**
   Reveals dependence or drift.

4. **Segments**
   Checks whether errors concentrate in groups.

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MSE, RMSE, MAE and R-squared

FORMULA / WORKED VALUE

$$MSE = \Sigma e^2/n$$
$$RMSE = \sqrt{MSE}$$
$$MAE = \Sigma|e|/n$$

Interpret the quantity, unit, population and assumptions before using the result.

1. **MSE:** Average squared error; strongly penalises large misses.
2. **RMSE:** Square root of MSE in outcome units.
3. **MAE:** Average absolute error; more robust to large misses.
4. **R^2:** Variation explained relative to a mean baseline.

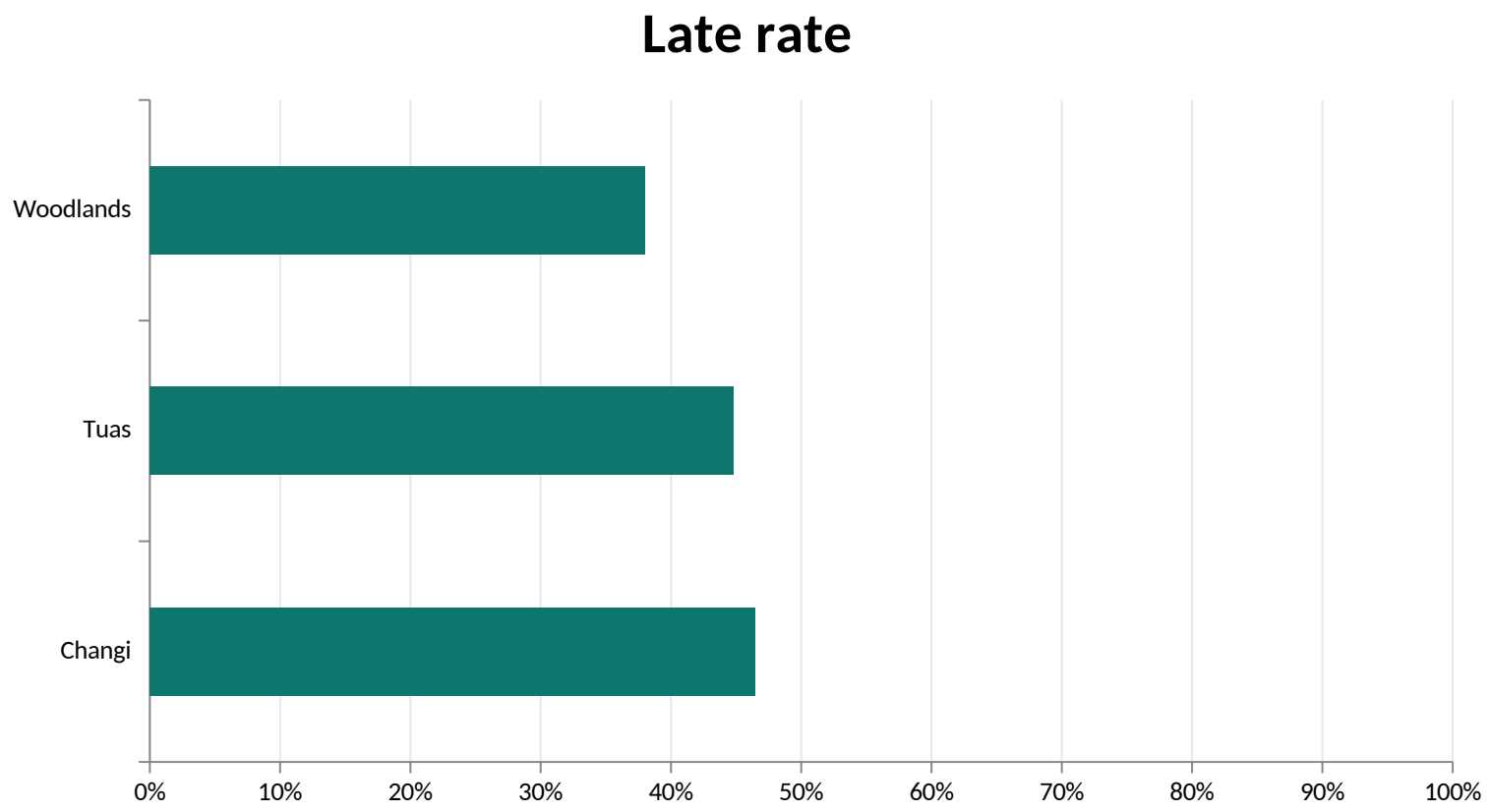
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Model credibility and communication

1. Data: Representative, valid and decision-time available.
2. Method: Assumptions match design and grain.
3. Evidence: Held-out metrics, diagnostics and sensitivity checks.
4. Claim: Action, guardrail and limitation stay within support.

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Late-delivery rate by warehouse



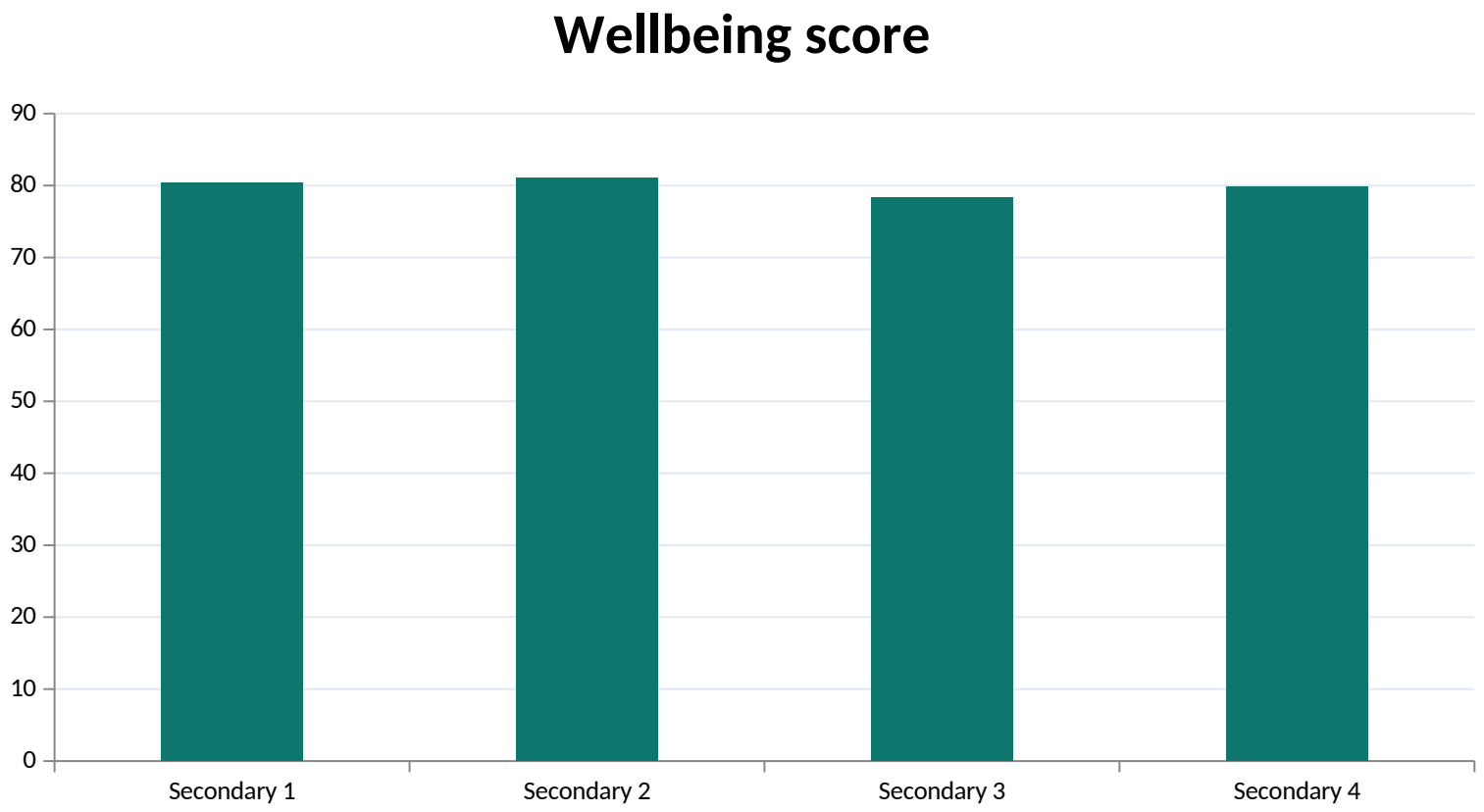
PYTHON → PPTX

A one-command pipeline reproduces the risk result.

Editable native PowerPoint chart derived from pandas output.

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Synthetic wellbeing by education level



PYTHON → PPTX

Synthetic model outputs are associations, not causal claims.

Editable native PowerPoint chart derived from pandas output.

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LAB 09 · CONCEPT BRIEF

Build a Reusable pandas Analytics Pipeline

Refactor a fulfilment analysis into small testable functions connected with pipe, assign and vectorised transformations.

YOU WILL BUILD

A reproducible analytics pipeline, validation log and late-delivery risk chart.

TOOLS Python, pandas pipe, assign, custom functions



ACCEPTANCE TEST

The pipeline can be re-run from raw input, validation fails loudly on broken schema, assertions pass and every output is derived without manual edits.

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LAB 10 · CONCEPT BRIEF

Capstone: Public Health Insight and Regression Review

Prepare, visualise and model synthetic student-health indicators, then review the credibility of the evidence for a programme decision.

YOU WILL BUILD

A complete capstone pack with clean data, summaries, visual dashboard, linear model, residual review and stakeholder brief.

TOOLS Python, pandas, Matplotlib, Seaborn, scikit-learn



ACCEPTANCE TEST

The script reproduces every output, test-set metrics and residuals are reported, causal claims are avoided and limitations are explicit in the stakeholder brief.

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TOPIC RECAP

Recap — Advanced Data Analytics

1. **Knowledge**
   Method chains and related principles.

2. **Ability**
   LO5

3. **Evidence**
   Runnable Python and pandas outputs.

4. **Next**
   Connect to the next analytical layer.

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WRAP-UP

Recap & Next Steps

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Course Feedback

1. **Share**
   Complete the course feedback form.

2. **Reflect**
   Note what worked well for you.

3. **Improve**
   Tell us what we can do better.

4. **Apply**
   Use one analysis at work this month.

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THANK YOU

Turn trusted data into a defensible decision.

Prepare, transform, visualise, analyse and review evidence with reproducible Python.

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